Denavit–Hartenberg Convention

The standard (distal) DH convention is used. The transform from frame $\{i-1\}$ to frame $\{i\}$ is

$$\begin{aligned}\mathbf{A}_i &= \mathrm{Rot}_z(\theta_i)\ \mathrm{Trans}_z(d_i)\ \mathrm{Trans}_x(a_i)\ \mathrm{Rot}_x(\alpha_i) \\ &= \begin{bmatrix} \cos\theta_i & -\sin\theta_i\cos\alpha_i & \sin\theta_i\sin\alpha_i & a_i\cos\theta_i \\ \sin\theta_i & \cos\theta_i\cos\alpha_i & -\cos\theta_i\sin\alpha_i & a_i\sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}.\end{aligned} \tag{1}$$

Table 1: DH parameters (q_i: joint angles of the MuJoCo model).

i	θ_i	d_i [m]	a_i [m]	α_i
1	$\pi - q_1$	0.0563	0	$\pi/2$
2	$q_2 + \gamma_2$	0	0.109307	0
3	$-q_3 + \gamma_3 - \gamma_2$	0	0.100516	0
4	$q_4 + \frac{\pi}{2} - \gamma_3$	−0.001253	0	$\pi/2$
5	$-q_5$	0.110	0	0

The constant joint offsets and the fixed base and tool transforms are

$$\gamma_2 = \mathrm{atan2}(0.1083,\ 0.0148) = 1.434980 \text{ rad}, \qquad {}^{W}\mathbf{T}_0 = \mathrm{Trans}(0.012,\ 0,\ 0)\,, \tag{2}$$

$$\gamma_3 = \mathrm{atan2}(0.0026999,\ 0.10048) = 0.026864 \text{ rad}, \qquad {}^{5}\mathbf{T}_{ee} = \mathrm{Trans}(0.0005,\ 0.0105,\ 0)\,. \tag{3}$$

Forward Kinematics

$$ {}^{W}\mathbf{T}_{ee}(\mathbf{q}) = {}^{W}\mathbf{T}_0\ \mathbf{A}_1\,\mathbf{A}_2\,\mathbf{A}_3\,\mathbf{A}_4\,\mathbf{A}_5\ {}^{5}\mathbf{T}_{ee} = \begin{bmatrix} \mathbf{R}(\mathbf{q}) & \mathbf{p}(\mathbf{q}) \\ \mathbf{0}^\top & 1 \end{bmatrix}, \tag{4}$$

where, with $C_i = \cos\theta_i$ and $S_i = \sin\theta_i$,

$$\mathbf{A}_1 = \begin{bmatrix} C_1 & 0 & S_1 & 0 \\ S_1 & 0 & -C_1 & 0 \\ 0 & 1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{A}_2 = \begin{bmatrix} C_2 & -S_2 & 0 & a_2C_2 \\ S_2 & C_2 & 0 & a_2S_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{A}_3 = \begin{bmatrix} C_3 & -S_3 & 0 & a_3C_3 \\ S_3 & C_3 & 0 & a_3S_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$\mathbf{A}_4 = \begin{bmatrix} C_4 & 0 & S_4 & 0 \\ S_4 & 0 & -C_4 & 0 \\ 0 & 1 & 0 & d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{A}_5 = \begin{bmatrix} C_5 & -S_5 & 0 & 0 \\ S_5 & C_5 & 0 & 0 \\ 0 & 0 & 1 & d_5 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Expanding Eq. (4), with $\psi = q_2 - q_3 + q_4$, $\tau_x = 0.0005\cos q_5 + 0.0105\sin q_5$ and $\tau_y = 0.0105\cos q_5 - 0.0005\sin q_5$,

$$\rho = a_2\cos(q_2+\gamma_2) + a_3\cos(q_2 - q_3 + \gamma_3) + d_5\cos\psi - \tau_x\sin\psi, \tag{5}$$

$$h = a_2\sin(q_2+\gamma_2) + a_3\sin(q_2 - q_3 + \gamma_3) + d_5\sin\psi + \tau_x\cos\psi, \tag{6}$$

$$\delta = d_4 + \tau_y, \tag{7}$$

the end-effector position is

$$\boxed{\begin{aligned} X &= 0.012 - \rho\cos q_1 + \delta\sin q_1, \\ Y &= \rho\sin q_1 + \delta\cos q_1, \\ Z &= d_1 + h. \end{aligned}} \tag{8}$$